Wei-Jhih Huang

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Research Interests

AI-Native Wireless Networks and Next-Generation Radio Access Networks, Deep Reinforcement Learning for Cross-Layer Resource Allocation and Optimization, Deep Generative Models for Out-of-Distribution Detection and Adaptive Learning

Education

Bachelor, Communication Engineering, National Central University ***Aug 2022 – Present***
GPA: 3.94/4.3, Expected Graduation: July 2026 Taoyuan, Taiwan

Research Experience

Information Processing and Communications Lab, National Central University ***Feb 2024 – Feb 2026***
Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang** Taoyuan, Taiwan
Keywords: Wireless Networking Systems, Resource Allocation, Deep Reinforcement Learning, Generative AI models

- Designed a learning-based RRM framework improving convergence and robustness for dynamic resource orchestration in high-mobility B5G systems, achieving ***67%*** faster convergence under dynamic user trajectories.
- Built a trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by ***90%*** and improving user quality of service by ***11%*** in cross-layer resource management scenarios.

Electronic Design Automation Lab, Chinese University of Hong Kong ***Jul – Aug 2025***
Summer Visiting Research Student, Advisor: **Prof. Bei Yu** Hong Kong
Keywords: Electronic Design Automation, Graph Learning, Graph Neural Network, Logic synthesis

- Studied NUA-Timer (Non-Uniform Arrival Timer), a BPN-based pre-synthesis timing prediction framework designed to capture long-range dependencies within complex circuit topologies under non-uniform input arrival times.
- Collaborated with the first author to extend the framework, contributing to Transformer-based architectural planning and loss function refinement through feasibility analysis of constrained logic synthesis in downstream EDA flows.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management." accepted to IEEE Wireless Communications and Networking Conference (WCNC), 2026.

Work Experience

Chunghwa Telecom ***Sep – Dec 2025***
AI Software Development Intern Taipei, Taiwan

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by ***33%.***

Industrial Technology Research Institute ***Jun – Sep 2024***
Summer AI Software Intern Hsinchu, Taiwan

- Implemented a hybrid edge-cloud architecture using Raspberry Pi and Cloud LLMs for a service robot at Taipei Songshan Sports Center; optimized backend processing to reduce latency by ***50%*** for 24/7 automated customer support.

Honors & Awards

- **Undergraduate Research Fellowship,** National Science and Technology Council (NSTC), Taiwan, 2025–2026
- **Honorable Mention,** Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Merit Award,** Individual Service-Learning Excellence Scholarship, National Central University, 2025
- **Excellence Award,** NCU Club Evaluation (as Presenter and President, NCUgF Rural Service Club), 2024
- **High Distinction and Public Service Scholarship,** National Central University (as President, NCUgF), 2024

Teaching Experience

National Central University ***Feb – Jun 2025***

Teaching Assistant, Courses: **Natural Language Processing** and **Enterprise Application of Pattern Recognition** Taoyuan, Taiwan

- Assisted in lecture preparation and grading for NLP topics for ***60*** students and image recognition topics for ***60*** students, analyzing the underlying AI technologies of real-world company products to support students in final project research.

Skills

AI Expertise: DRL (DDPG, PPO, Meta-RL), Deep Generative Models (VAE, Diffusion Model), GNN, LLMs

AI Application: System design and deployment (Linux-based database in C, OpenAI API integration), agentic workflow automation (React, n8n, RAG with intent routing), LLM-based chatbot development (Azure OpenAI, Linux backend services)

Programming: C/C++, Python (PyTorch), JavaScript (Node.js, Express), Linux environments

Tools & Frameworks: Git, Docker, Azure, OpenAI API, n8n, React, REST APIs

Languages: English (TOEFL 89, TOEIC 810), Mandarin Chinese (Native)

Leadership & Extracurriculars

NCU Rural Service Club (NCUgF) ***Feb 2023 – Jul 2025***

Club President & Service Team's Group Leader

- Secured multiple awards and scholarships for the club through service initiatives as club president (see Honors & Awards).
- Led and coordinated educational camps for ***160*** children across remote elementary schools in Yilan in February 2023 and Taichung in January 2024, designing interactive lessons and managing team activities to foster creativity and learning.

NCU Wireless Communication and Network Multimedia Camp ***Jul 2023***

AI Instructor & Course Team Member

- Created course materials and delivered lectures on machine learning and AI fundamentals to ***70*** high school students.
- Guided students in completing hands-on Arduino projects to foster practical understanding of wireless communication.